

Linear System of ODE

A system of n -linear ODE in n -unknowns:

$$\begin{array}{ccccccc} P_{11}(D)x_1 & + & P_{12}(D)x_2 & + & \cdots & + & \cdots & + & P_{1n}(D)x_n & = & b_1(t) \\ P_{21}(D)x_1 & + & P_{22}(D)x_2 & + & \cdots & + & \cdots & + & P_{2n}(D)x_n & = & b_2(t) \\ & & \cdots & & \cdots & & \cdots & & \cdots & & \\ P_{n1}(D)x_1 & + & P_{n2}(D)x_2 & + & \cdots & + & \cdots & + & P_{nn}(D)x_n & = & b_n(t) \end{array}$$

where the P_{ij} are polynomials of various degrees in the differential operator D .

The system of first-order ODEs in normal form (a special case of the above system) is,

$$\begin{array}{l} \frac{d}{dt}(x_1) = g_1(t, x_1, x_2, \dots, x_n) = a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t) \\ \frac{d}{dt}(x_2) = g_2(t, x_1, x_2, \dots, x_n) = a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t) \\ \vdots \\ \frac{d}{dt}(x_n) = g_n(t, x_1, x_2, \dots, x_n) = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t) \end{array}$$

If all $f_i(t) = 0$, $i = 1, 2, \dots, n$ then the above system is said to be **homogeneous**; otherwise the system is called **non-homogeneous**.

Linear System of ODE

Example 01: Homogeneous Form

$$\begin{aligned} \frac{d}{dt}(x) &= 3x + 4y \\ \frac{d}{dt}(y) &= 5x - 7y \end{aligned} \Rightarrow \begin{aligned} D(x) &= 3x + 4y \\ D(y) &= 5x - 7y \end{aligned} \Rightarrow \begin{aligned} (D - 3)x - 4y &= 0 \\ 5x - (D + 7)y &= 0 \end{aligned}$$

Example 02: Non-homogeneous Form

$$\begin{aligned} \frac{d}{dt}(x) &= 6x + y + z + t \\ \frac{d}{dt}(y) &= 8x + 7y - z + 10t \\ \frac{d}{dt}(z) &= 2x + 9y - z + 6t \end{aligned} \Rightarrow \begin{aligned} D(x) &= 6x + y + z + t \\ D(y) &= 8x + 7y - z + 10t \\ D(z) &= 2x + 9y - z + 6t \end{aligned} \Rightarrow \begin{aligned} (D - 6)x - y - z &= t \\ -8x + (D - 7)y + z &= 10t \\ -2x - 9y + (D + 1)z &= 6t \end{aligned}$$

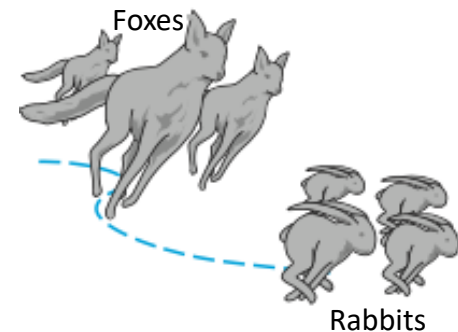
- Predator-Prey Population Model:**

$$\begin{aligned} \frac{dP_1}{dt} &= aP_1 - bP_1P_2 \\ \frac{dP_2}{dt} &= -mP_2 + nP_1P_2 \end{aligned}$$

where P_1 : population of rabbits, P_2 : population of foxes

a : growth rate of rabbits, b : killing rate of rabbits,

m : death rate of foxes, and n : growth rate of foxes.



Linear System of ODE: Homogeneous Form

Solution of Homogeneous system of ODE by Elimination

Example 01:

$$\begin{aligned}\frac{d}{dt}(x) &= 3x + 4y &\Rightarrow D(x) &= 3x + 4y &\Rightarrow (D - 3)x - 4y &= 0 &\quad (1) \\ \frac{d}{dt}(y) &= 5x - 7y &\Rightarrow D(y) &= 5x - 7y &\Rightarrow 5x - (D + 7)y &= 0 &\quad (2)\end{aligned}$$

Solution. Multiplying Eq. (1) by 5 and operating Eq. (2) by $(D - 3)$ and then subtracting eliminates x from the given system, i.e.,

$$5(D - 3)x - 20y = 0 \quad (1)$$

$$5(D - 3)x - (D - 3)(D + 7)y = 0 \quad (2)$$

Now, Eq. (1) - Eq. (2) $\Rightarrow$

$$(D - 3)(D + 7)y - 20y = 0 \Rightarrow \{(D + 7)(D - 3) - 20\}y = 0 \Rightarrow (D^2 + 4D - 41)y = 0$$

From which, the auxiliary roots are, $m = -2 \pm 3\sqrt{5} \Rightarrow m_1 = -2 + 3\sqrt{5}, m_2 = -2 - 3\sqrt{5}$

$$\text{Thus, } y(t) = c_1 e^{(-2+3\sqrt{5})t} + c_2 e^{(-2-3\sqrt{5})t} = e^{-2t} (c_1 e^{3\sqrt{5}t} + c_2 e^{-3\sqrt{5}t})$$

Linear System of ODE: Homogeneous Form

Solution of Homogeneous system of ODE by Elimination

Example 01:

$$\begin{aligned} \frac{d}{dt}(x) &= 3x + 4y & \Rightarrow D(x) &= 3x + 4y & \Rightarrow (D - 3)x - 4y &= 0 & (1) \\ \frac{d}{dt}(y) &= 5x - 7y & \Rightarrow D(y) &= 5x - 7y & \Rightarrow 5x - (D + 7)y &= 0 & (2) \end{aligned}$$

Solution. Now substituting the solution of $y(t)$ into Eq. (1) we obtain,

$$\begin{aligned} x(t) &= \frac{4}{D - 3} y = \frac{4}{D - 3} \left[c_1 e^{(-2+3\sqrt{5})t} + c_2 e^{(-2-3\sqrt{5})t} \right] \\ &= 4 \left[\frac{c_1}{(-2 + 3\sqrt{5}) - 3} e^{(-2+3\sqrt{5})t} + \frac{c_2}{(-2 - 3\sqrt{5}) - 3} e^{(-2-3\sqrt{5})t} \right] \\ &= 4 \left[\frac{c_1}{-5 + 3\sqrt{5}} e^{(-2+3\sqrt{5})t} + \frac{c_2}{-5 - 3\sqrt{5}} e^{(-2-3\sqrt{5})t} \right]. \end{aligned}$$

Linear System of ODE: Homogeneous Form

Solution of Homogeneous system of ODE by Elimination

Example 02:

$$Dx + (D + 2)y = 0 \quad (1)$$

$$(D - 3)x - 2y = 0 \quad (2)$$

Solution. Operating Eq. (1) by $(D - 3)$ and Eq. (2) by D and then subtracting eliminates x from the given system, i.e.,

$$D(D - 3)x + (D + 2)(D - 3)y = 0 \quad (1)$$

$$D(D - 3)x - 2Dy = 0 \quad (2)$$

Now, Eq. (1) - Eq. (2) $\Rightarrow$

$$(D - 3)(D + 2)y + 2Dy = 0 \Rightarrow \{(D + 2)(D - 3) + 2D\}y = 0 \Rightarrow (D^2 + D - 6)y = 0$$

From which, the auxiliary equation is $m^2 + m - 6 = 0 \Rightarrow (m + 3)(m - 2) = 0$

and the auxiliary roots are, $m = -3, 2$

Thus, $y(t) = c_1 e^{-3t} + c_2 e^{2t}$

Linear System of ODE: Homogeneous Form

Solution of Homogeneous system of ODE by Elimination

Example 02:

$$Dx + (D + 2)y = 0 \quad (1)$$

$$(D - 3)x - 2y = 0 \quad (2)$$

Solution. Now substituting the solution of $y(t)$ into Eq. (2) we obtain,

$$x(t) = \frac{2}{D - 3} y = \frac{2}{D - 3} [c_1 e^{-3t} + c_2 e^{2t}]$$

$$= 2 \left[\frac{c_1}{(-3) - 3} e^{-3t} + \frac{c_2}{(2) - 3} e^{2t} \right]$$

$$= -\frac{1}{3} c_1 e^{-3t} - 2c_2 e^{2t}.$$

Linear System of ODE: Non-Homogeneous Form

Solution of Non-homogeneous system of ODE by Elimination

Example 03:

$$\begin{aligned}\frac{dx}{dt} &= -y + t & \Rightarrow & Dx + y = t & (1) \\ \frac{dy}{dt} &= x - t & & x - Dy = t & (2)\end{aligned}$$

Solution. Operating Eq. (2) by D and Eq. (1) by 1 and then subtracting eliminates x from the given system, i.e.,

$$Dx + y = t \quad (1)$$

$$Dx - D^2y = 1 \quad (2)$$

Now, Eq. (1) – Eq. (2) $\Rightarrow$

$$D^2y + y = t - 1 \Rightarrow (D^2 + 1)y = t - 1$$

From which, the auxiliary equation is $m^2 + 1 = 0 \Rightarrow m = \pm i$

Thus, the complementary solution for $y(t)$ is, $y_c(t) = c_1 \cos t + c_2 \sin t$.

The particular solution for $y(t)$, is

$$y_p(t) = \frac{1}{1 + D^2}(t - 1) = (1 + D^2)^{-1}(t - 1) = (1 - D^2)(t - 1) = (t - 1)$$

Linear System of ODE: Non-Homogeneous Form

Solution of Non-homogeneous system of ODE by Elimination

Example 03:

$$\begin{aligned}\frac{dx}{dt} &= -y + t & \Rightarrow & Dx + y = t & (1) \\ \frac{dy}{dt} &= x - t & & x - Dy = t & (2)\end{aligned}$$

Solution. Therefore, the solution of $y(t)$ becomes,

$$y(t) = y_c(t) + y_p(t) = c_1 \cos t + c_2 \sin t + t - 1$$

Now substituting the solution of $y(t)$ into Eq. (2) we obtain,

$$x(t) = t + Dy = -c_1 \sin t + c_2 \cos t + t + 1.$$

Linear System of ODE: Non-Homogeneous Form

Solution of Non-homogeneous system of ODE by Elimination

Example 04:

$$D^2x - 4y = e^t \quad (1)$$

$$4x - D^2y = e^t \quad (2)$$

Solution. Operating Eq. (1) by D^2 and multiplying Eq. (2) by 4 and then subtracting eliminates y from the given system, i.e.,

$$D^4x - 4D^2y = e^t \quad (1)$$

$$16x - 4D^2y = 4e^t \quad (2)$$

Now, Eq. (1) – Eq. (2) $\Rightarrow$

$$D^4x - 16x = -3e^t \Rightarrow (D^4 - 16)x = -3e^t$$

From which, the auxiliary equation is $m^4 - 16 = 0 \Rightarrow (m^2 + 4)(m^2 - 4) = 0$

And the auxiliary roots are, $m = \pm 2, \pm 2i$.

Thus, the complementary solution for $x(t)$ is,

$$x_c(t) = c_1 e^{2t} + c_2 e^{-2t} + c_3 \cos 2t + c_4 \sin 2t.$$

The particular solution for $x(t)$, is

$$x_p(t) = -3 \frac{1}{D^4 - 16} (e^t) = -\frac{3}{1^4 - 16} (e^t) = \frac{1}{5} e^t$$

Linear System of ODE: Non-Homogeneous Form

Solution of Non-homogeneous system of ODE by Elimination

Example 04:

$$D^2x - 4y = e^t \quad (1)$$

$$4x - D^2y = e^t \quad (2)$$

Solution. Therefore, the solution of $x(t)$ becomes,

$$x(t) = x_c(t) + x_p(t) = c_1 e^{2t} + c_2 e^{-2t} + c_3 \cos 2t + c_4 \sin 2t + \frac{1}{5} e^t$$

Now substituting the solution of $y(t)$ into Eq. (1) we obtain,

$$y(t) = \frac{1}{4} (D^2x - e^t) = \frac{1}{4} \left(4c_1 e^{2t} + 4c_2 e^{-2t} - 4c_3 \cos 2t - 4c_4 \sin 2t + \frac{1}{5} e^t - e^t \right)$$

$$= c_1 e^{2t} + c_2 e^{-2t} - c_3 \cos 2t - c_4 \sin 2t - \frac{1}{5} e^t.$$

Linear System of ODE

Exercise 4.9

Solve the following system of differential equations by systematic elimination.

1. $\frac{dx}{dt} = 2x - y$

2. $\frac{dx}{dt} = 4x + 7y$

10. $D^2x - Dy = t$
 $(D + 3)x + (D + 3)y = 2$

$\frac{dy}{dt} = x$

$\frac{dy}{dt} = x - 2y$

11. $(D^2 - 1)x - y = 0$
 $(D - 1)x + Dy = 0$

3. $\frac{dx}{dt} = -y + t$

4. $\frac{dx}{dt} - 4y = 1$

12. $(2D^2 - D - 1)x - (2D + 1)y = 1$
 $(D - 1)x + Dy = -1$

$\frac{dy}{dt} = x - t$

$\frac{dy}{dt} + x = 2$

13. $2\frac{dx}{dt} - 5x + \frac{dy}{dt} = e^t$

$\frac{dx}{dt} - x + \frac{dy}{dt} = 5e^t$

5. $(D^2 + 5)x - 2y = 0$
 $-2x + (D^2 + 2)y = 0$

14. $\frac{dx}{dt} + \frac{dy}{dt} = e^t$

6. $(D + 1)x + (D - 1)y = 2$
 $3x + (D + 2)y = -1$

$-\frac{d^2x}{dt^2} + \frac{dx}{dt} + x + y = 0$

7. $\frac{d^2x}{dt^2} = 4y + e^t$

8. $\frac{d^2x}{dt^2} + \frac{dy}{dt} = -5x$

15. $(D - 1)x + (D^2 + 1)y = 1$
 $(D^2 - 1)x + (D + 1)y = 2$

$\frac{d^2y}{dt^2} = 4x - e^t$

$\frac{dx}{dt} + \frac{dy}{dt} = -x + 4y$

16. $D^2x - 2(D^2 + D)y = \sin t$
 $x + Dy = 0$

9. $Dx + D^2y = e^{3t}$
 $(D + 1)x + (D - 1)y = 4e^{3t}$